

How Honeybees Decide Where to Live

Every spring, roughly half of a healthy honeybee colony leaves home. The old queen departs with about 10,000 workers, and the swarm clusters on a nearby branch while scouts search for a new nest. Within two or three days, the swarm chooses one site, lifts off together, and flies to it. The decision is remarkably good: swarms almost always pick a cavity of the right size, high off the ground, with a small, defensible entrance.

Thomas Seeley, a biologist at Cornell University, spent decades working out how a group of insects with brains the size of a grass seed makes such a decision without any leader. His experiments on Appledore Island, off the coast of Maine, used swarms placed on an island with no natural cavities, so the only options were nest boxes that Seeley's team built and placed.

The Scouts Report Back

A few hundred scouts, mostly older foragers, leave the cluster and inspect possible homes. When a scout finds a promising cavity, she returns and performs a waggle dance on the surface of the swarm. The dance encodes the direction and distance of the site. Its length encodes how good the scout thinks the site is: a scout who found an excellent cavity may dance more than 100 circuits, while a scout who found a mediocre one dances only a handful.

Other scouts that watch the dance fly out to inspect the site themselves. If they agree, they come back and dance for it too. Crucially, each scout's enthusiasm fades with every return trip, so a site keeps its support only if new scouts keep being recruited. Poor sites fade out. Good sites build a growing crowd of dancers.

A Quorum, Not a Majority

For years, researchers assumed the swarm decided when a majority of dancers favored one site. In 2004, Seeley and Kirk Visscher showed that the trigger is different. Scouts at the chosen site count how many other scouts are present there. When roughly 15 to 20 scouts are at the same cavity at the same time, they begin a new behavior called piping: a high-pitched buzz produced by pressing the thorax against other bees. Piping tells the whole swarm to warm up its flight muscles. About an hour later, when the cluster's temperature reaches 35 degrees Celsius, the swarm takes off.

This quorum rule has two advantages. It is fast, because the swarm does not have to wait for every scout to agree. And it is accurate, because reaching a quorum at a site requires many independent inspections, which filters out a single scout's mistake.

What the Bees Are Measuring

Scouts evaluate several features of a cavity. In Seeley's experiments, swarms preferred a volume of about 40 liters, an entrance of roughly 15 square centimeters, an entrance near the bottom of the cavity, and a height of at least 3 meters above the ground. A scout measures volume by walking the inside walls for 30 to 60 minutes, covering a total path of several meters. She returns to the cavity several times, and each inspection appears to refine her estimate.

The swarm does not need any bee to know all of this. The quality of a site is spread across many short dances, each one a small vote, and the dances compete for the attention of uncommitted scouts.

Lessons for People

Seeley has argued that honeybee swarms behave like a well-run committee. The scouts share a goal, they gather information independently before reporting, they advertise options in proportion to their quality, and they stop the debate as soon as enough evidence has accumulated. In his 2010 book *Honeybee Democracy*, he lists five habits that the bees follow and that human groups often skip: keep members' interests aligned, minimize the leader's influence, seek diverse solutions, aggregate knowledge through debate, and use a quorum for speed and accuracy.

The parallel is not perfect. Bees have no politics, and every scout is a sister of every other. But the core result stands on its own: a group of 10,000 animals, none of which has seen more than a few of the options, reliably chooses the best home in about 48 hours.